# Breakout



Register/login at https://scratch.mit.edu

*You have a bat and ball and have to knock out the bricks.*

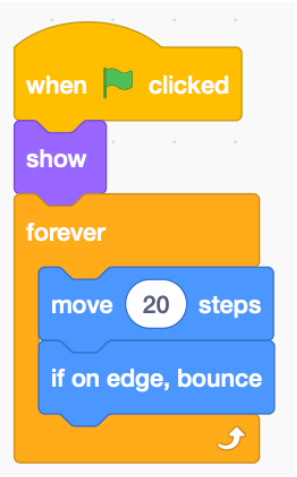
1) Make the bat. Create a new sprite and use **paint** to draw a rectangle in the centre.
2) Use the mouse to control the bat. Move the bat to the bottom of the screen and add code that **sets the x** (horizontal) position to **mouse x** in a loop.
3) Make a ball. Create a new sprite and choose the **Ball** sprite.
4) Add code to make the ball bounce around the screen. Make sure the ball is **show**ing and then **move** the ball inside a loop. Add "**if on edge, bounce**" to bounce off the edges.
5) **Hide** the ball if it gets past the bat. Add an **if** block in the loop to detect when the vertical (y) position of the ball nears the edge, before it bounces. Experiment with the y position.






Bat Code






Ball Code




Ball Code

6) When the ball is **touching** the **bat** it has to bounce off. Add an **if** block that changes the ball **direction**, subtracting it from 180°
(180 degrees is half a circle).



Ball Code

7) Give the player more control over the bounce by adding the horizontal (x) difference between the ball and the mouse, to the direction.

Ball Code



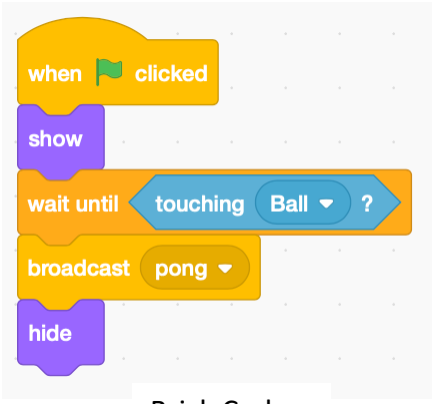

Add this inside the **if** block above.

8) Make a brick. Create a new colourful square sprite.

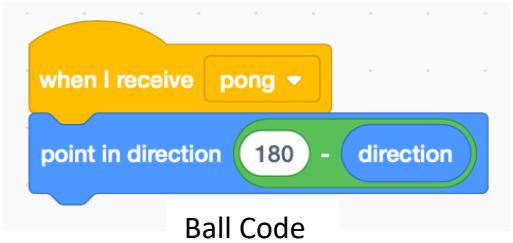


9) The brick will disappear when the ball hits it. **Wait until touching Ball**.

10) To make the ball bounce off the brick, **broadcast** a 'pong' message to the ball.



Brick Code



Ball Code

11) When the ball **receives pong** it bounces.

*Click on the green flag to restart the game.*

- *Duplicate more bricks and put them where you like.*
- *Try adding a score.*

***Save** your code with a good name. **File > Save now***